ByteTrack

Phan tich Toan hoc Chi tiet

Baseline (Kalman Filter) vs ByteTrack + LTC

Source code:

`yolox/tracker/kalman_filter.py`
`yolox/tracker/ltc_motion.py`
`yolox/tracker/byte_tracker.py`

June 25, 2026

Contents

1 Tong quan kien truc **2**

2 ByteTrack Baseline — Kalman Filter **2**

2.1 Khong gian trang thai (State Space) 2

2.2 Ma tran chuyen trang thai — Constant Velocity Model 2

2.3 Predict Step 3

2.4 Update Step 3

2.5 Khoi tao Track 4

2.6 Mahalanobis Gating Distance 4

2.7 Two-Stage Association — `byte_tracker.py:262-291` 5

2.7.1 Vong 1 — Detections score cao ($s > \tau_{\text{track}}$) 5

2.7.2 Vong 2 — Detections score thap ($0.1 < s < \tau_{\text{track}}$) 5

3 ByteTrack + LTC — Liquid Time-Constant Motion Predictor **6**

3.1 Dong luc (Motivation) 6

3.2 Feature Vector — Lich su quy dao 6

3.3 CfC Cell — Closed-form Continuous-time 6

3.3.1 Phuong trinh vi phan lien tuc 6

3.3.2 Nghiem giai tich (Closed-form Solution) — `ltc_motion.py:17-23` 6

3.4 Kien truc LtcMotionResidual — `ltc_motion.py:26-76` 7

3.4.1 Forward Pass qua 2 lop CfC 7

3.5 Residual Refinement — `ltc_motion.py:128-161` 7

3.5.1 Denormalize residual 7

3.5.2 Variance tu log 7

3.5.3 Hieu chinh mean va covariance 7

3.6 Muc tieu huan luyen (Training Objective) 8

4 So sanh Tong hop **9**

4.1 Ket luan 9

5 Bang Ky hieu **9**

1. Tong quan kien truc

ByteTrack la thuat toan multi-object tracking (MOT) su dung **tat ca** detection boxes (ke ca confidence thap) de duy tri tracklet. Pipeline moi frame t:

1. Du doan vi tri tuong lai qua **motion model** (Kalman hoac Kalman + LTC).
2. Ghep noi tracks $\leftrightarrow$ detections bang **Hungarian algorithm**.
3. Cap nhat trang thai Kalman.
4. Quan ly vong doi: *Tracked* $\rightarrow$ *Lost* $\rightarrow$ *Removed*.

2. ByteTrack Baseline — Kalman Filter

2.1. Khong gian trang thai (State Space)

$$\boldsymbol{x} = \underbrace{\begin{bmatrix} x \\ y \\ a \\ h \end{bmatrix}}_{\text{vi tri}} \oplus \underbrace{\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{a} \\ \dot{h} \end{bmatrix}}_{\text{van toc}} \in \mathbb{R}^8$$

(x, y)	Toa do tam bounding box (pixel)
$a = w/h$	Aspect ratio (chieu rong / chieu cao)
h	Chieu cao bounding box (pixel)
$(\dot{x}, \dot{y}, \dot{a}, \dot{h})$	Van toc cua tung thanh phan

Observation (quan sat duoc tu detector):

$$\boldsymbol{z}_t = \begin{bmatrix} x & y & a & h \end{bmatrix}^\top \in \mathbb{R}^4$$

Van toc **khong quan sat truc tiep** — phai uoc luong qua Kalman.

2.2. Ma tran chuyen trang thai — Constant Velocity Model

$$\mathbf{F} = \begin{array}{c} \\ x \\ y \\ a \\ h \\ \dot{x} \\ \dot{y} \\ \dot{a} \\ \dot{h} \end{array} \begin{array}{c} \begin{array}{cccccccc} x & y & a & h & \dot{x} & \dot{y} & \dot{a} & \dot{h} \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array} = \begin{bmatrix} \mathbf{I}_4 & \mathbf{I}_4 \\ \mathbf{0}_4 & \mathbf{I}_4 \end{bmatrix}$$

Y nghia — constant velocity:

$$x_t = x_{t-1} + \dot{x}_{t-1}, \qquad \dot{x}_t = \dot{x}_{t-1}$$

(tuong tu cho y, a, h). Ma tran quan sat $\mathbf{H} \in \mathbb{R}^{4\times 8}$:

$$\mathbf{H} = \begin{bmatrix} \mathbf{I}_4 & \mathbf{0}_4 \end{bmatrix} \quad \Rightarrow \quad \boldsymbol{z}_t = \mathbf{H}\,\boldsymbol{x}_t + \text{noise}$$

2.3. Predict Step

$$\hat{\boldsymbol{x}}_{t|t-1} = \mathbf{F}\,\hat{\boldsymbol{x}}_{t-1|t-1} \tag{1}$$

$$\mathbf{P}_{t|t-1} = \mathbf{F}\,\mathbf{P}_{t-1|t-1}\,\mathbf{F}^\top + \mathbf{Q}_t \tag{2}$$

Ma tran nhieu qua trinh $\mathbf{Q}_t$ — adaptive theo chieu cao h (`kalman_filter.py:107-117`):

$$\mathbf{Q}_t = \operatorname{diag}\left(\underbrace{\frac{h}{20}, \frac{h}{20}, 10^{-2}, \frac{h}{20}}_{\text{nhieu vi tri}}, \; \underbrace{\frac{h}{160}, \frac{h}{160}, 10^{-5}, \frac{h}{160}}_{\text{nhieu van toc}} \right)^{\odot 2}$$

Tai sao scale theo h?

Doi tuong lon (bounding box cao) di chuyen nhieu pixel hon. Scale theo h giup Kalman tu dieu chinh cho moi kich thuoc doi tuong.

2.4. Update Step

Buoc 1 — Chieu xuong measurement space:

$$\hat{\boldsymbol{z}}_t = \mathbf{H}\,\hat{\boldsymbol{x}}_{t|t-1} \tag{3}$$

$$\mathbf{S}_t = \mathbf{H}\,\mathbf{P}_{t|t-1}\,\mathbf{H}^\top + \mathbf{R}_t \tag{4}$$

Buoc 2 — Kalman Gain (dung Cholesky decomposition):

$$\mathbf{K}_t \;=\; \mathbf{P}_{t|t-1}\,\mathbf{H}^\top\,\mathbf{S}_t^{-1} \tag{5}$$

Buoc 3 — Hieu chinh:

$$\hat{\boldsymbol{x}}_{t|t} = \hat{\boldsymbol{x}}_{t|t-1} + \mathbf{K}_t\,\underbrace{(\boldsymbol{z}_t - \hat{\boldsymbol{z}}_t)}_{\text{innovation}} \tag{6}$$

$$\mathbf{P}_{t|t} = \mathbf{P}_{t|t-1} - \mathbf{K}_t\,\mathbf{S}_t\,\mathbf{K}_t^\top \tag{7}$$

Ma tran nhieu quan sat $\mathbf{R}_t$:

$$\mathbf{R}_t = \operatorname{diag}\left(\frac{h}{20},\ \frac{h}{20},\ 10^{-1},\ \frac{h}{20}\right)^{\odot 2}$$

Y nghia cua Kalman Gain $\mathbf{K}_t$

$\mathbf{K}_t$ can bang giua **prediction** va **observation**:

- $\mathbf{K}_t \to \mathbf{0}$: Tin vao prediction (observation nhieu lon)
- $\mathbf{K}_t \to \mathbf{H}^{-1}$: Tin vao observation (prediction kem)

2.5. Khoi tao Track

Khi detection chua duoc ghep voi track nao:

$$\hat{\boldsymbol{x}}_0 = \begin{bmatrix} \boldsymbol{z}_0 \\ \mathbf{0}_4 \end{bmatrix}, \qquad \mathbf{P}_0 = \operatorname{diag}(2\sigma_p h,\ \ldots,\ 10\sigma_v h,\ \ldots)^2$$

Van toc khoi tao $= \mathbf{0}$, covariance van toc lon ($\times 10$) vi chua biet toc do.

2.6. Mahalanobis Gating Distance

$$d_M(\boldsymbol{z},\,\hat{\boldsymbol{x}}) \;=\; \left(\boldsymbol{z} - \mathbf{H}\hat{\boldsymbol{x}}\right)^\top \mathbf{S}^{-1} \left(\boldsymbol{z} - \mathbf{H}\hat{\boldsymbol{x}}\right) \;\le\; \chi^2_{4,\,0.95} = 9.4877$$

Nguong lay tu phan phoi chi-square voi 4 bac tu do o muc tin cay 95%.

2.7. Two-Stage Association — `byte_tracker.py:262-291`

2.7.1. Vong 1 — Detections score cao ($s > \tau_{\text{track}}$)

1. **IoU Distance:** $d_{ij}^{\text{IoU}} = 1 - \text{IoU}(\hat{b}_i, b_j)$
2. **Score Fusion:** $d_{ij} = d_{ij}^{\text{IoU}} \cdot (1 - s_j)$
3. **Hungarian Algorithm** voi nguong $\tau_{\text{match}} = 0.8$

2.7.2. Vong 2 — Detections score thap ($0.1 < s < \tau_{\text{track}}$)

Chi dung IoU distance thuan, nguong $= 0.5$, cho cac tracks chua ghep.

Y tuong ByteTrack

Detections score thap thuong la doi tuong bi che khuat. Khong loai bo ngay giup duy tri tracklet qua occlusion.

3. ByteTrack + LTC — Liquid Time-Constant Motion Predictor

3.1. Dong luc (Motivation)

Van de	Nguyen nhan
Khong mo hinh hoa chuyen dong phi tuyen	Gia dinh constant velocity
Nhay cam voi occlusion keo dai	Khong dung lich su dai han
Δt co dinh $= 1$	Khong xu ly frame gap khong deu
Uncertainty co dinh theo h	Khong hoc duoc tu du lieu

Giai phap: Huan luyen mang neural (CfC) de hoc **residual** giua Kalman prediction va vi tri thuc te tu lich su quy dao.

3.2. Feature Vector — Lich su quy dao

$$\boldsymbol{f}_s = \Big[\underbrace{x,\ y,\ a,\ h,\ \dot{x},\ \dot{y},\ \dot{a},\ \dot{h}}_{\text{Kalman state (8 chieu)}},\ \underbrace{\Delta t}_{\substack{\text{frame}\\\text{gap}}},\ \underbrace{\mathbf{1}_{\text{miss}}}_{\substack{\text{dang}\\\text{bi mat}}},\ \underbrace{c_{\text{miss}}}_{\substack{\text{so frame}\\\text{mat lien tiep}}},\ \underbrace{s_t}_{\substack{\text{detection}\\\text{confidence}}}\Big]$$

Lich su 16 buoc gan nhat: $\mathcal{H}_t = \{\boldsymbol{f}_{t-15}, \ldots, \boldsymbol{f}_t\} \in \mathbb{R}^{16\times 12}$

3.3. CfC Cell — Closed-form Continuous-time

3.3.1. Phuong trinh vi phan lien tuc

$$\tau \frac{d\boldsymbol{h}}{dt} = -\alpha \odot \boldsymbol{h}(t) + f\big(\boldsymbol{x}(t), \boldsymbol{h}(t)\big)$$

$\boldsymbol{h}(t) \in \mathbb{R}^H$	Hidden state tai thoi diem t
$\tau \in \mathbb{R}^H_{>0}$	**Time constant** — hang so thoi gian cua moi neuron
$\alpha \in \mathbb{R}^H_{>0}$	**Damping coefficient** — he so tat dan
$f(\cdot)$	Candidate mapping (phi tuyen)

3.3.2. Nghiem giai tich (Closed-form Solution) — `ltc_motion.py:17-23`

Voi buoc thoi gian Δt, f duoc xap xi hang so:

$$\boxed{\begin{aligned} \tilde{\boldsymbol{h}}_t &= \tanh\Big(\mathbf{W}_x\,\boldsymbol{x}_t + \mathbf{W}_h\,\boldsymbol{h}_{t-1}\Big) && \text{(candidate state)} \quad (8)\\ \boldsymbol{g}_t &= \exp\left(-\frac{\alpha \odot \Delta t}{\tau}\right) && \text{(time-aware decay gate)} \quad (9)\\ \boldsymbol{h}_t &= \boldsymbol{g}_t \odot \boldsymbol{h}_{t-1} \;+\; (1-\boldsymbol{g}_t)\odot \tilde{\boldsymbol{h}}_t && \text{(hidden state update)} \quad (10) \end{aligned}}$$

$\alpha = \text{softplus}(\hat{\alpha}) + \epsilon$ va $\tau = \text{softplus}(\hat{\tau}) + \epsilon$ la cac **tham so hoc duoc per-neuron**.

Y nghia cua gate g_t

Dieu kien	**Gia tri g_t**	**Hanh vi**
$\Delta t \to 0$	$\boldsymbol{g}_t \to \mathbf{1}$	Trang thai hau nhu khong doi
$\Delta t \to \infty$	$\boldsymbol{g}_t \to \mathbf{0}$	Hoan toan thay bang candidate
Δt binh thuong	$\boldsymbol{g}_t \in (0,1)$	To hop co trong so

Moi neuron co τ rieng $\to$ hoc duoc **time scale khac nhau**: neuron nhanh (τ nho) theo doi chuyen dong tuc thoi, neuron cham (τ lon) nam bat xu huong dai han.

3.4. Kien truc LtcMotionResidual — `ltc_motion.py:26-76`

3.4.1. Forward Pass qua 2 lop CfC

Buoc 0 — Chuan hoa dau vao:

$$\hat{\mathcal{H}} = \text{LayerNorm}(\mathcal{H}), \qquad \Delta t_s = \mathcal{H}_{s,8}$$

Buoc 1 — Chay tuan tu qua $T = 16$ buoc, $L = 2$ lop:

$$\boldsymbol{h}_s^{(0)} = \text{CfC}^{(0)}\Big(\hat{\boldsymbol{f}}_s,\ \boldsymbol{h}_{s-1}^{(0)},\ \Delta t_s\Big) \tag{11}$$

$$\boldsymbol{h}_s^{(1)} = \text{CfC}^{(1)}\Big(\boldsymbol{h}_s^{(0)},\ \boldsymbol{h}_{s-1}^{(1)},\ \Delta t_s\Big) \tag{12}$$

Khoi tao: $\boldsymbol{h}_0^{(l)} = \mathbf{0}$.

Buoc 2 — Output head:

$$\mathbb{R}^H \xrightarrow{\text{LayerNorm}} \mathbb{R}^H \xrightarrow{\text{Linear}} \mathbb{R}^H \xrightarrow{\text{GELU}} \mathbb{R}^H \xrightarrow{\text{Linear}} \mathbb{R}^8 \quad \Rightarrow \quad \begin{bmatrix} \boldsymbol{r} \\ \log \boldsymbol{v} \end{bmatrix}, \quad \boldsymbol{r}, \log \boldsymbol{v} \in \mathbb{R}^4$$

3.5. Residual Refinement — `ltc_motion.py:128-161`

3.5.1. Denormalize residual

$$\boldsymbol{r}_{\text{final}} = \boldsymbol{r} \odot \boldsymbol{\sigma}^* + \boldsymbol{\mu}^* \qquad \boldsymbol{r}_{\text{clipped}} = \text{clip}(\boldsymbol{r}_{\text{final}},\ -256,\ +256)$$

3.5.2. Variance tu log

$$\boldsymbol{v} = \exp\Big(\text{clip}(\log \boldsymbol{v},\ -10,\ +10)\Big) \odot (\boldsymbol{\sigma}^*)^{\odot 2}$$

3.5.3. Hieu chinh mean va covariance

Hieu chinh mean (chi 4 thanh phan vi tri):

$$\hat{\boldsymbol{x}}_{t|t-1}^{\text{refined}} = \hat{\boldsymbol{x}}_{t|t-1} + \begin{bmatrix} \boldsymbol{r}_{\text{clipped}} \\ \mathbf{0}_4 \end{bmatrix}$$

Hieu chinh covariance (tang uncertainty khi model khong chac):

$$\mathbf{P}_{t|t-1}^{\text{refined}} = \mathbf{P}_{t|t-1} + \lambda_{\text{cov}} \cdot \text{diag}(\boldsymbol{v},\ \mathbf{0}_4)$$

voi λ_{cov} = `covariance_scale`.

Co che tu dieu chinh

Khi model khong chac ve residual ($\boldsymbol{v}$ lon), covariance $\mathbf{P}$ tang $\Rightarrow$ Kalman gain $\mathbf{K}$ tang $\Rightarrow$ Update step **tin vao observation hon prediction**.

3.6. Muc tieu huan luyen (Training Objective)

Model du doan **residual giua Kalman prediction va vi tri thuc**:

$$\boldsymbol{\delta}^* = \boldsymbol{x}_{\text{true}} - \hat{\boldsymbol{x}}_{\text{Kalman}}$$

$$\mathcal{L} = \frac{1}{2} \sum_{i=1}^{4} \left[\log v_i + \frac{(r_i - \delta_i^*)^2}{v_i} \right]$$

Ham loss nay dong thoi:

- Toi thieu hoa sai so du doan $(r_i - \delta_i^*)^2$.
- Calibrate variance v_i: neu kho du doan, model tang v_i (trung thuc ve uncertainty).

4. So sanh Tong hop

Khia canh	Kalman Baseline	ByteTrack + LTC
Motion model	Constant velocity (tuyen tinh)	CfC ODE (phi tuyen, hoc tu data)
Xu ly thoi gian	$\Delta t = 1$ co dinh	Δt bien doi qua time constant τ
Lich su su dung	Khong (Markov)	16 buoc $\times$ 12 features
Occlusion	Van toc ve 0 khi *Lost*	Feature `is_missing`, `missing_count`
Uncertainty	Co dinh theo h	Variance $\boldsymbol{v}$ adaptive (hoc duoc)
Dau ra	$\hat{\boldsymbol{x}}$, $\mathbf{P}$	$\hat{\boldsymbol{x}} + \boldsymbol{r}$, $\mathbf{P} + \lambda\mathbf{V}$
Khi nao ap dung	Luc nao cung	Khi track co $\geq$ `min_history` buoc

Table 1: So sanh Kalman Filter baseline va ByteTrack + LTC

4.1. Ket luan

Diem mau chot

LTC **khong thay the** Kalman Filter — no la mot **post-hoc corrector** chay sau predict step. Kalman van dam nhan tinh toan uncertainty (covariance), con LTC bo sung kien thuc ve trajectory patterns ma Gaussian noise model khong nam bat duoc.

5. Bang Ky hieu

Tai lieu duoc tao tu phan tich source code project ByteTrack + LTC.

Ky hieu	Y nghia
$\boldsymbol{x}_t \in \mathbb{R}^8$	Vector trang thai Kalman
$\boldsymbol{z}_t \in \mathbb{R}^4$	Observation tu detector
$\mathbf{F}$	Ma tran chuyen trang thai
$\mathbf{H}$	Ma tran quan sat
$\mathbf{Q}_t$, $\mathbf{R}_t$	Nhieu qua trinh / nhieu quan sat
$\mathbf{P}_t$	Covariance matrix
$\mathbf{K}_t$	Kalman Gain
d_M	Mahalanobis distance (gating)
$\boldsymbol{h}_t^{(l)} \in \mathbb{R}^H$	Hidden state lop l cua CfC
τ, $\alpha \in \mathbb{R}^H$	Time constant va damping (learned)
$\boldsymbol{g}_t \in (0,1)^H$	Time-aware decay gate
$\boldsymbol{r} \in \mathbb{R}^4$	Predicted residual tu LTC
$\boldsymbol{v} \in \mathbb{R}^4$	Predicted variance tu LTC
$\mathcal{H}_t \in \mathbb{R}^{16\times 12}$	Lich su quy dao 16 buoc
λ_{cov}	Covariance scale hyperparameter
$\odot$	Hadamard product (nhan theo phan tu)
$\chi^2_{k,p}$	Nguong chi-square k bac tu do, muc p

Table 2: Bang ky hieu su dung trong tai lieu